

| Assessment | Test | Domain  | Skill                             | Difficulty                               |
|------------|------|---------|-----------------------------------|------------------------------------------|
| SAT        | Math | Algebra | Linear equations in two variables | Hard <div></div> <div></div> <div></div> |

Question

$\frac{3}{5}x + \frac{3}{4}y = 7$

Which table gives three values of  $x$  and their corresponding values of  $y$  for the given equation?

Answer

A.

| $x$ | $y$              |
|-----|------------------|
| 1   | $\frac{113}{20}$ |
| 2   | $\frac{101}{20}$ |
| 4   | $\frac{77}{20}$  |

B.

| $x$ | $y$            |
|-----|----------------|
| 1   | $\frac{47}{5}$ |
| 2   | $\frac{44}{5}$ |
| 4   | $\frac{38}{5}$ |

C.

| $x$ | $y$              |
|-----|------------------|
| 1   | $\frac{148}{15}$ |
| 2   | $\frac{136}{15}$ |
| 4   | $\frac{112}{15}$ |

D.

| $x$ | $y$              |
|-----|------------------|
| 1   | $\frac{128}{15}$ |
| 2   | $\frac{116}{15}$ |
| 4   | $\frac{92}{15}$  |

Correct Answer: D

Rationale

Choice D is correct. Each of the tables gives the same three values of  $x$ : 1, 2, and 4. Substituting 1 for  $x$  in the given equation yields  $(\frac{3}{5})(1) + \frac{3}{4}y = 7$ , or  $\frac{3}{5} + \frac{3}{4}y = \frac{35}{5}$ . Subtracting  $\frac{3}{5}$  from both sides of this equation yields  $\frac{3}{4}y = \frac{32}{5}$ . Multiplying both sides of this equation by  $\frac{4}{3}$  yields  $y = \frac{128}{15}$ . Therefore, when  $x = 1$ , the corresponding value of  $y$  for the given equation is  $\frac{128}{15}$ . Substituting 2 for  $x$  in the given equation yields  $(\frac{3}{5})(2) + \frac{3}{4}y = 7$ , or  $\frac{6}{5} + \frac{3}{4}y = \frac{35}{5}$ . Subtracting  $\frac{6}{5}$  from both sides of this equation yields  $\frac{3}{4}y = \frac{29}{5}$ . Multiplying both sides of this equation by  $\frac{4}{3}$  yields  $y = \frac{116}{15}$ . Therefore, when  $x = 2$ , the corresponding value of  $y$  for the given equation is  $\frac{116}{15}$ . Substituting 4 for  $x$  in the given equation yields  $(\frac{3}{5})(4) + \frac{3}{4}y = 7$ , or  $\frac{12}{5} + \frac{3}{4}y = \frac{35}{5}$ . Subtracting  $\frac{12}{5}$  from both sides of this equation yields  $\frac{3}{4}y = \frac{23}{5}$ . Multiplying both sides of this equation by  $\frac{4}{3}$  yields  $y = \frac{92}{15}$ . Therefore, when  $x = 4$ , the corresponding value of  $y$  for the given equation is  $\frac{92}{15}$ . The table in choice

D gives  $x$ -values of **1**, **2**, and **4** and corresponding  $y$ -values of  $\frac{128}{15}$ ,  $\frac{116}{15}$ , and  $\frac{92}{15}$ , respectively. Therefore, the table in choice D gives three values of  $x$  and their corresponding values of  $y$  for the given equation.

Choice A is incorrect. This table gives three values of  $x$  and their corresponding values of  $y$  for the equation  $\frac{3}{5}x + \frac{3}{4}y = 7$ .

Choice B is incorrect. This table gives three values of  $x$  and their corresponding values of  $y$  for the equation  $\frac{3}{5}x + y = 10$ .

Choice C is incorrect. This table gives three values of  $x$  and their corresponding values of  $y$  for the equation  $\frac{3}{5}x + \frac{3}{4}y = 8$ .